Unit IV

4. Domain name Disputes and Cyber Crimes

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4.1 Background of Domain Names:

Domain names were introduced to make the **Internet user-friendly** by replacing complex numerical Internet Protocol (IP) addresses with easy-to-remember names. In the early days of the Internet, computers communicated only using IP addresses, which were difficult for humans to recall.

Key Points:

1. **Need for Domain Names:**
   Every device on the Internet has a unique **IP address** (e.g., 192.168.1.1). Domain names were created to map these numbers to meaningful names like **www.example.com**.

2. **Domain Name System (DNS):**
   The **Domain Name System (DNS)** was developed as a hierarchical and distributed naming system that translates domain names into corresponding IP addresses.

3. **Evolution of Domain Structure:**
   Domain names are structured in levels, such as **Top-Level Domains (TLDs)** (.com, .org, edu), **Country Code TLDs** (.in, .uk), and **Second-Level Domains** (example).

4. **Global Management:**
   The Internet Corporation for Assigned Names and Numbers (**ICANN**) was established to manage domain name allocation and ensure global uniqueness.

5. **Legal and Commercial Importance:**
   With the growth of **e-commerce and online services**, domain names gained commercial value, leading to legal issues like **trademark conflicts and cybersquatting**.

In short:

Domain names were developed to simplify Internet access by converting numeric IP addresses into meaningful names, forming a crucial part of Internet communication and online identity.

4.2 Where lies the dispute?

The dispute in domain names mainly lies in the **conflict between domain name registration and trademark rights**. Since domain names are unique and registered on a 'first-come, first-served basis', legal issues arise when they clash with existing brand names or trademarks.

Key Points:

1. **Domain Name vs Trademark Conflict:**
   A dispute arises when a domain name is identical or confusingly similar to a registered trademark, causing confusion among users.

2. **Cybersquatting:**
   Disputes often occur when a person registers a well-known trademark as a domain name with the intention of selling it to the rightful owner at a high price.

3. **Bad Faith Registration:**
   If a domain name is registered to mislead customers, damage reputation, or divert traffic for commercial gain, it leads to legal disputes.

4. **Jurisdictional Issues:**
   Domain names operate globally, while trademark laws are territorial, creating conflicts over which country's laws apply.

5. **Consumer Confusion and Brand Dilution:**
   Disputes arise when users are misled into believing a domain belongs to a genuine brand, harming business goodwill and consumer trust.

In short:

The dispute lies in the **unauthorized or bad-faith use of domain names that infringe trademark rights**, leading to confusion, misuse, and legal conflict.

4.3 Insertion of Internet Domain Names and the trademark Law:

1. Insertion of Internet Domain Names:

The insertion of Internet domain names refers to the **registration and inclusion of domain names in the Domain Name System (DNS)**, which gives an entity an online identity. Domain names are registered on a **first-come, first-served basis**, without initial verification of trademark rights.

Key Points:

1. Domain names are inserted into the **DNS database** through accredited registrars.
2. Registration does not automatically check for **existing trademarks**, leading to conflicts.
3. A registered domain name provides **global visibility and online identity**.
4. Issues arise when domain names are similar to **well-known brand names**.
5. This system has led to disputes such as **cybersquatting and passing off**.

In short:

Insertion of domain names creates online identity but can result in legal disputes when it conflicts with trademark rights.

2. Trademark Law:

Trademark law protects symbols, names, logos, and brand identities that distinguish goods or services of one entity from another. In India, trademarks are governed by the **Trade Marks Act, 1999**.

Key Points:

1. Trademarks protect **brand identity and goodwill** of businesses.
2. Rights are generally **territorial**, unlike domain names which are global.

3. Trademark law prevents **unauthorized use** of identical or confusingly similar marks.
4. Domain names are treated as **business identifiers**, similar to trademarks.
5. Trademark owners can take legal action against **infringing domain names**.

In short:

Trademark law safeguards brand identity and is used to resolve disputes where domain names infringe registered trademarks.

4.4 Classification of Cyber Crime, Target of computer crime:

1. Classification of Cyber Crime:

Cyber crimes are unlawful acts committed using computers, networks, or the internet. They can be classified based on the **target or victim** involved.

Types of Cyber Crime:

1. **Crimes Against Individuals:**
   Includes cyber stalking, online harassment, identity theft, phishing, email spoofing, and defamation.

2. **Crimes Against Property:**
   Involves damage to digital property such as data theft, hacking, malware attacks, software piracy, and intellectual property theft.

3. **Crimes Against Organizations:**
   Includes unauthorized access to corporate systems, data breaches, denial of service (DoS) attacks, and industrial espionage.

4. **Crimes Against Society:**
   Involves spreading obscene content, fake news, hate speech, online gambling, and circulation of illegal materials.

5. **Crimes Against Government:**
   Includes cyber terrorism, hacking government websites, espionage, and attacks on critical infrastructure.

In short:

Cyber crimes are classified based on the **victim affected**, ranging from individuals to governments.

2. Target of Computer Crime:

The target of computer crime refers to the **entities or resources attacked** by cyber criminals.

Main Targets:

1. **Individuals:**
   Personal data, bank details, passwords, and social media accounts are common targets.

2. **Businesses and Organizations:**
   Corporate data, customer databases, trade secrets, and financial systems are targeted for financial gain.

3. **Government Systems:**
   Government databases, defense networks, and public service systems are targeted for political or terrorist motives.

4. **Computer Systems and Networks:**
   Attackers target servers, websites, and networks to disrupt services or steal information.

5. **Digital Assets and Information:**
   Includes intellectual property, software, confidential data, and digital currency.

In short:
Computer crimes target **people, data, systems, organizations, and governments**, aiming to steal, damage, or disrupt digital resources.

4.5 Damage to computer System: Unauthorized Access, Packet Sniffing, Tempest attack, Password Cracking, Butter overflow:

1. Unauthorized Access:

Unauthorized access refers to accessing a computer system, network, or data **without permission** of the owner. It is a serious cyber offence under cyber laws.

Key Points:

1. Occurs when an attacker bypasses security controls like login credentials or firewalls.
2. Can be done by hacking passwords, exploiting software vulnerabilities, or using stolen credentials.
3. Leads to **data theft, data modification, or system damage**.
4. Violates privacy and confidentiality of users.
5. Punishable under the **Information Technology Act, 2000**.

In short:
Unauthorized access damages computer systems by allowing illegal entry and misuse of data.

2. Packet Sniffing:

Packet sniffing is a technique where an attacker secretly **captures and monitors data packets** traveling over a network.

Key Points:

1. Attackers use sniffing tools to intercept network traffic.
2. Sensitive data like usernames, passwords, and bank details can be stolen.
3. Common in unsecured or public Wi-Fi networks.
4. Can lead to identity theft and financial fraud.
5. Prevented using encryption protocols like **HTTPS and VPNs**.

In short:
Packet sniffing damages computer systems by **stealing confidential data during transmission**.

3. TEMPEST Attack:

A **TEMPEST attack** is a sophisticated cyber attack where attackers capture **electromagnetic signals** emitted by electronic devices.

Key Points:

1. Computers emit electromagnetic radiation during operation.
2. Attackers intercept these signals to reconstruct data.
3. No physical access to the system is required.
4. Used mainly in **espionage and military cyber attacks**.
5. Prevented using shielding and secure hardware.

In short:

TEMPEST attacks compromise systems by leaking data through electromagnetic emissions.

4. Password Cracking:

Password cracking is the process of illegally obtaining passwords to gain unauthorized access to systems.

Key Points:

1. Uses methods like brute force, dictionary attacks, and keylogging.
2. Weak or simple passwords are easy targets.
3. Enables attackers to access private accounts and data.
4. Can lead to identity theft and system misuse.
5. Prevented by strong passwords and two-factor authentication.

In short:

Password cracking damages computer systems by breaking authentication security.

5. Buffer Overflow (Butter Overflow):

Buffer overflow is a programming error where more data is written into memory than it can handle.

Key Points:

1. Occurs due to poor input validation in software.
2. Excess data overwrites adjacent memory locations.
3. Attackers can execute malicious code.
4. Can cause system crashes or unauthorized control.
5. Prevented by secure coding practices.

In short:

Buffer overflow attacks damage computer systems by exploiting software weaknesses to gain control.

4.6 Computer virus: Viruses, Logic Bomb, Worms, Trojan Horse Programme, Denial of Service, Tampering with Computer Source Documents:

1. Computer Virus:

A **computer virus** is a malicious program that attaches itself to legitimate files and spreads when the

infected file is executed.

Key Points:

1. Requires a **host program** to operate and spread.
2. Replicates itself and infects other files or systems.
3. Can corrupt data, slow down systems, or crash computers.
4. Spreads through email attachments, USB drives, and downloads.
5. Prevented using updated **antivirus software**.

In short:

A virus damages systems by infecting files and spreading malicious code.

2. Logic Bomb:

A **logic bomb** is malicious code that activates when a **specific condition or time** is met.

Key Points:

1. Remains dormant until triggered.
2. Trigger may be a date, event, or action.
3. Used to delete files or damage systems.
4. Often planted by insiders or disgruntled employees.
5. Difficult to detect before activation.

In short:

Logic bombs silently damage systems when preset conditions are fulfilled.

3. Worms:

A **worm** is a standalone malicious program that **replicates itself** without human intervention.

Key Points:

1. Does not require a host program.
2. Spreads automatically through networks.
3. Consumes system resources and network bandwidth.
4. Can slow down or crash servers.
5. Prevented by firewalls and security patches.

In short:

Worms damage networks by self-replication and resource exhaustion.

4. Trojan Horse Programme:

A **Trojan Horse** is a malicious program disguised as a legitimate or useful application.

Key Points:

1. Appears harmless but contains hidden malware.
2. Does not self-replicate like a virus.
3. Creates backdoors for attackers.
4. Used to steal data or gain unauthorized access.
5. Spread through fake software downloads and emails.

In short:

Trojan programs deceive users and allow attackers to control systems.

5. Denial of Service (DoS) Attack:

A **Denial of Service (DoS)** attack aims to make a computer or network **unavailable to users**.

Key Points:

1. Floods the system with excessive requests.
2. Overloads servers and crashes services.
3. Distributed DoS (DDoS) uses multiple systems.
4. Commonly targets websites and online services.
5. Causes financial and service disruption.

In short:

DoS attacks damage systems by interrupting normal services.

6. Tampering with Computer Source Documents:

Tampering with computer source documents refers to **unauthorized modification or destruction of source code**.

Key Points:

1. Includes altering, deleting, or hiding source code.
2. Affects software functionality and security.
3. Often done to introduce malicious code.
4. Considered a serious offence under **IT Act, 2000 (Section 65)**.
5. Can cause system failure or data loss.

In short:

Tampering with source documents damages computer systems and software integrity.